Business Statement

Customer Churn Analysis and Prediction for a Telecom Company

Tools: Python, pandas, seaborn, scikit-learn

1. Business Problem

A telecom provider is losing a large share of its customers. In the dataset analysed, 1,869 of 7,043 customers (26.5%) churned. Together these customers account for about $139,000 in monthly recurring billing, roughly 30% of all monthly charges, or about $1.67 million per year. Acquiring a new customer typically costs more than retaining an existing one, so unexplained churn is an expensive problem.

The company does not currently know which customers are most likely to leave, or which factors drive their decision, so retention spending cannot be targeted.

2. Objective

- **Understand:** identify the customer, contract, billing and service factors most associated with churn.
- **Predict:** build a machine learning model that flags customers likely to churn so the retention team can act early.
- **Recommend:** translate the findings into practical retention actions.

3. Data and Approach

The analysis uses the Telco Customer Churn dataset (7,043 customers). The target variable is Churn (Yes/No). The work followed these stages:

- **Data cleaning:** converted TotalCharges from text to numeric, handled 11 blank values (new customers with zero tenure), and checked for missing values and duplicates.
- **Exploratory analysis:** compared churn rates across contract, payment method, internet service, tenure and charges.
- **Modelling:** one-hot encoded categorical features and trained Logistic Regression, Decision Tree and Random Forest models on an 80/20 stratified split, using tenure, monthly charges, total charges, contract type and payment method as predictors.
- **Optimisation:** addressed class imbalance with balanced class weights, tuned hyperparameters with GridSearchCV, and tuned the decision threshold to favour catching churners.

4. Key Insights

Factor	Finding
Contract type	Month-to-month customers churn at 42.7%, versus 11.3% on one-year and 2.8% on two-year contracts. This is the strongest driver of churn.
Tenure	Churners stay about 18 months on average versus about 38 months for retained customers. Churn is concentrated among newer customers.
Payment method	Electronic check users churn at 45.3%, compared with 15-19% for all other payment methods, including both automatic options.
Internet service	Fiber optic customers churn at 41.9% versus 19.0% for DSL and 7.4% for customers with no internet service.
Monthly charges	Churners pay more on average ($74 versus $61 per month), which suggests price sensitivity among higher-paying customers.

5. Model Results

Because only about one in four customers churns, accuracy alone is misleading. A model can look accurate while missing most churners. The business goal is to catch customers who are about to leave, so churn recall and F1 were prioritised.

Model	Accuracy	Churn Recall	Churn F1
Logistic Regression (baseline)	78%	48%	54%
Decision Tree	72%	45%	46%
Random Forest	77%	49%	53%
Balanced + Tuned Logistic Regression	73%	79%	61%
Final: Balanced Logistic Regression, threshold 0.55	**75%**	**76%**	**62%**

The final model, a balanced Logistic Regression with a 0.55 threshold, correctly identified **284 of the 374 churners (76%)** in the test set, compared with under 50% for the baseline models. The trade-off is that about half of the customers it flags (265 of 549) would not have left, so retention offers should be low-cost and well targeted.

6. Recommendations

- **Move customers to longer contracts.** Offer discounts or perks to month-to-month customers who switch to one- or two-year plans.
- **Promote automatic payments.** Encourage electronic check users to move to bank transfer or credit card autopay.
- **Focus on early-tenure customers.** Introduce onboarding, check-ins and offers during the first months, when churn risk is highest.
- **Review the fiber optic experience.** Investigate pricing, reliability and support for fiber customers, whose churn rate is more than double that of DSL.
- **Deploy the model for targeted retention.** Score customers regularly and direct retention offers to the highest-risk group instead of the whole base.

7. Business Value

This project shows how analytics can turn a broad retention problem into a targeted action plan. By focusing on the customers most likely to leave, the company can reduce revenue loss, spend its retention budget more efficiently, and improve customer lifetime value. Each percentage point of churn avoided protects recurring revenue from the highest-risk segments.

Limitations: Results are based on a single historical dataset and a limited set of features; financial figures describe billing associated with churned customers and are not a forecast of savings. Future work could add more features, cross-validated model comparison and a cost-benefit analysis of retention offers.